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```python
import kagglehub

# Dataset download
path = kagglehub.dataset_download("rxnach/student-stress-factors-a-comprehensive-analysis")
print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/rxnach/student-stress-factors-a-comprehensive-a
100%|██████████| 11.7k/11.7k [00:00<00:00, 12.8MB/s]Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/rxnach/student-stress-factors-a-compreh
```

```python
import os
print(os.listdir(path))
```

```
['StressLevelDataset.csv']
```

```python
import pandas as pd

# Path and file name ni kalipi load chesthunnam
file_path = f"{path}/StressLevelDataset.csv"
df = pd.read_csv(file_path)

# Ippudu columns list kanipisthundi
print("Columns in Dataset:")
print(df.columns)

# Data samples chudadaniki
df.head()
```

```
Columns in Dataset:
Index(['anxiety_level', 'self_esteem', 'mental_health_history', 'depression',
       'headache', 'blood_pressure', 'sleep_quality', 'breathing_problem',
       'noise_level', 'living_conditions', 'safety', 'basic_needs',
       'academic_performance', 'study_load', 'teacher_student_relationship',
       'future_career_concerns', 'social_support', 'peer_pressure',
       'extracurricular_activities', 'bullying', 'stress_level'],
      dtype='object')
```

	anxiety_level	self_esteem	mental_health_history	depression	headache	blood_pressure	sle
0	14	20	0	11	2	1	
1	15	8	1	15	5	3	
2	12	18	1	14	2	1	
3	16	12	1	15	4	3	
4	16	28	0	7	2	3	

5 rows × 21 columns

```python
# 1. Null values check (Missing values)
print("\nMissing values:")
print(df.isnull().sum())

# 2. Duplicates remove cheyadam
df = df.drop_duplicates()

# 3. Basic stats chudadaniki
df.describe()
```

```
Missing values:
anxiety_level                   0
self_esteem                     0
mental_health_history           0
depression                      0
headache                        0
blood_pressure                  0
sleep_quality                   0
breathing_problem               0
noise_level                     0
living_conditions               0
safety                          0
basic_needs                     0
academic_performance            0
study_load                      0
teacher_student_relationship    0
future_career_concerns          0
social_support                  0
peer_pressure                   0
extracurricular_activities      0
bullying                        0
stress_level                    0
dtype: int64
```

	anxiety_level	self_esteem	mental_health_history	depression	headache	blood_pressure
count	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000

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```
       'noise_level', 'living_conditions', 'safety', 'basic_needs',
       'academic_performance', 'study_load', 'teacher_student_relationship',
       'future_career_concerns', 'social_support', 'peer_pressure',
       'extracurricular_activities', 'bullying', 'stress_level'],
      dtype='object')
```

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basic_needs                     0
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study_load                      0
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social_support                  0
peer_pressure                   0
extracurricular_activities      0
bullying                        0
stress_level                    0
dtype: int64
```

	anxiety_level	self_esteem	mental_health_history	depression	headache	blood_pressure
count	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000	1100.000000
mean	11.063636	17.777273	0.492727	12.555455	2.508182	2.181818
std	6.117558	8.944599	0.500175	7.727008	1.409356	0.833575
min	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000
25%	6.000000	11.000000	0.000000	6.000000	1.000000	1.000000
50%	11.000000	19.000000	0.000000	12.000000	3.000000	2.000000
75%	16.000000	26.000000	1.000000	19.000000	3.000000	3.000000
max	21.000000	30.000000	1.000000	27.000000	5.000000	3.000000

8 rows × 21 columns

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

# 1. Selecting limited features for a better UI experience
features = ['sleep_quality', 'study_load', 'academic_performance',
            'extracurricular_activities', 'social_support']
X = df[features]
y = df['stress_level']

# 2. Splitting data (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Training the Model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# 4. Checking Accuracy
y_pred = model.predict(X_test)
```

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dtype: int64

	anxiety_level	self_esteem	mental_health_history	depression	headache	blood_pressure
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model.fit(X_train, y_train)

# 4. Checking Accuracy
y_pred = model.predict(X_test)
print(f"Model Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%")
```
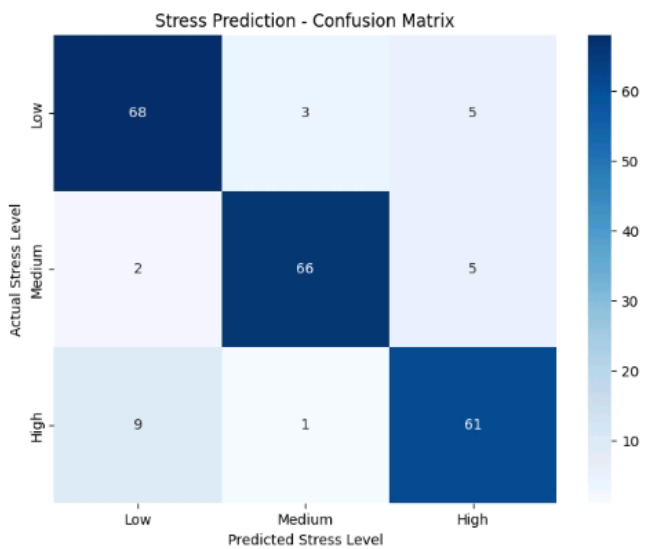

```
Model Accuracy: 88.64%
```

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

# Confusion Matrix create cheyadam
cm = confusion_matrix(y_test, y_pred)

# Heatmap drawing
plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Low', 'Medium', 'High'],
            yticklabels=['Low', 'Medium', 'High'])
plt.xlabel('Predicted Stress Level')
plt.ylabel('Actual Stress Level')
plt.title('Stress Prediction - Confusion Matrix')
plt.show()
```



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